

Probability Output — Percentile Calculation Line by Line

This PDF explains exactly how the Percentile column is produced from the 15 Sales observations.

1. Sales data

Sales values are already in ascending order: 105, 112, 120, 128, 135, 141, 150, 158, 165, 171, 180, 188, 195, 202, 212.

Number of observations:

$$n = 15$$

2. The easy percentile formula

For the i -th value in the sorted data, Excel's Probability Output percentile rank is:

$$\text{Percentile} = [(i - 0.5) / n] \times 100$$

Here $n = 15$, so:

$$\text{Percentile} = [(i - 0.5) / 15] \times 100$$

3. Why the first percentile is 3.333333

First observation: $i = 1$, Sales = 105

$$\text{Percentile} = [(1 - 0.5) / 15] \times 100$$

$$= (0.5 / 15) \times 100$$

$$= 0.0333333 \times 100$$

$$= 3.333333\%$$

4. Why the second percentile is 10

Second observation: $i = 2$, Sales = 112

$$\text{Percentile} = [(2 - 0.5) / 15] \times 100$$

$$= (1.5 / 15) \times 100$$

$$= 10\%$$

5. All 15 percentile calculations

i	Sales	i – 0.5	(i – 0.5) / 15	× 100	Percentile
1	105	0.5	0.03333333	× 100	3.33333333
2	112	1.5	0.10000000	× 100	10.00000000
3	120	2.5	0.16666667	× 100	16.66666667
4	128	3.5	0.23333333	× 100	23.33333333
5	135	4.5	0.30000000	× 100	30.00000000
6	141	5.5	0.36666667	× 100	36.66666667
7	150	6.5	0.43333333	× 100	43.33333333
8	158	7.5	0.50000000	× 100	50.00000000
9	165	8.5	0.56666667	× 100	56.66666667
10	171	9.5	0.63333333	× 100	63.33333333
11	180	10.5	0.70000000	× 100	70.00000000
12	188	11.5	0.76666667	× 100	76.66666667
13	195	12.5	0.83333333	× 100	83.33333333
14	202	13.5	0.90000000	× 100	90.00000000
15	212	14.5	0.96666667	× 100	96.66666667

6. Line-by-line examples

Observation 1: Sales = 105

$$\text{Percentile} = [(1 - 0.5) / 15] \times 100$$

$$= (0.5 / 15) \times 100$$

$$= 3.33333333\%$$

Observation 3: Sales = 120

$$\text{Percentile} = [(3 - 0.5) / 15] \times 100$$

$$= (2.5 / 15) \times 100$$

$$= 16.66666667\%$$

Observation 8: Sales = 158

$$\text{Percentile} = [(8 - 0.5) / 15] \times 100$$

$$= (7.5 / 15) \times 100$$

$$= 50.00000000\%$$

Observation 10: Sales = 171

$$\text{Percentile} = [(10 - 0.5) / 15] \times 100$$

$$= (9.5 / 15) \times 100$$

$$= 63.33333333\%$$

Observation 15: Sales = 212

$$\text{Percentile} = [(15 - 0.5) / 15] \times 100$$

$$= (14.5 / 15) \times 100$$

$$= 96.66666667\%$$

7. Final Probability Output

Percentile	Sales
3.33333333	105
10.00000000	112
16.66666667	120
23.33333333	128
30.00000000	135
36.66666667	141
43.33333333	150
50.00000000	158
56.66666667	165
63.33333333	171
70.00000000	180
76.66666667	188
83.33333333	195
90.00000000	202
96.66666667	212

8. Easy way to remember

Step 1 → Sort the Sales values from smallest to largest.

Step 2 → Count the observations: $n = 15$.

Step 3 → Give each value a position $i = 1, 2, 3, \dots, 15$.

Step 4 → Use Percentile = $[(i - 0.5) / n] \times 100$.

Step 5 → Substitute $n = 15$.

Step 6 → The middle observation ($i = 8$) has percentile 50%, so Sales = 158 is the 50th percentile in this output.

9. Important point

This Probability Output is assigning a percentile rank to each observed Sales value. It is not calculating new Sales values by interpolation. That is why each row keeps the original Sales value and receives a percentile.